

Contextual Write–Read–Value Attention: An Architecture-Path Study

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Abstract

We study contextual Write–Read–Value (W/R/V) attention in approximately 98.2M-parameter causal language models. W/R/V uses eight contextual read heads, two shared contextual write/value heads, per-head RMS normalization, rotary position encoding, and causal softmax through PyTorch scaled dot-product attention. In a controlled H200 comparison, GQA and W/R/V use the same tokenizer, FineWeb-Edu stream, 100,007,936-token budget, optimizer, peak learning rate, BF16 numerical mode, evaluation protocol, and three seeds. W/R/V improves held-out loss on all three paired seeds: its mean is 4.684432 ± 0.021137 versus 4.703035 ± 0.017657 for GQA, with a mean paired difference of -0.018604 . Mean training throughput is 121,448 versus 125,021 tokens/s, a 2.86% penalty. With only three pairs, the 95% t interval for the mean loss difference, $[-0.0541, 0.0169]$, includes zero; we therefore report a consistent small-scale observation rather than universal superiority. Historical fixed-state WVR runs and lexical-write variants are retained only as architecture-path evidence. The controlled W/R/V condition disables the lexical write residual, isolating the contextual attention path used in the main comparison.

1 Introduction

Attention-based language models combine contextual transport with content-dependent selection over individual past token representations. We study a grouped contextual Write–Read–Value (W/R/V) parameterization and its quality–throughput tradeoff against grouped-query attention (GQA) under a matched small-model budget.

Every token traverses the same dense computation path. The main sequence mixer is causal W/R/V attention with eight contextual read heads and two shared contextual write/value heads. Per-head RMS normalization and rotary position encoding precede PyTorch scaled dot-product attention. Both W/R/V and GQA retain the same tied lexical-object feed-forward branches and deterministic micro-experts; the controlled W/R/V runs disable the separate lexical write residual.

The architecture emerged from a systematic fixed-state WVR study. Those recurrent variants are consistent with a limitation of a single compressed matrix for instance-selective retrieval. We retain them as negative path ablations rather than presenting them as the final architecture.

Our contributions are empirical:

- contextual W/R/V attention with grouped read/write heads, per-head RMS normalization, RoPE, and SDPA configured to permit FlashAttention;
- a controlled three-seed H200 comparison against GQA at 100,007,936 training tokens per run;
- a mean W/R/V held-out loss of 4.684432 versus 4.703035 for GQA, with W/R/V lower on all three paired seeds and 2.86% lower training throughput;
- fixed-state WVR and attention-path ablations that motivate, but do not isolate, hypotheses about compressed and lexical addressing; and

- raw-artifact-backed reporting that separates completed observations from optimizer-matched architectural claims.

The paired result is consistent but small: with three seeds, its 95% interval includes zero. We therefore claim neither statistical significance nor universal architectural superiority. The contribution is a fully archived matched observation and an executable architecture path that can be tested at larger scale.

2 Method

2.1 Contextual W/R/V attention

Let $\mathbf{H} \in \mathbb{R}^{B \times T \times d}$ be normalized hidden states. Each layer projects contextual reads, writes, and values,

$$\mathbf{R} = \mathbf{H}\mathbf{W}_r, \quad \mathbf{W}_{\text{ctx}} = \mathbf{H}\mathbf{W}_w, \quad \mathbf{V} = \mathbf{H}\mathbf{W}_v. \quad (1)$$

The evaluated model uses eight read heads and two shared write/value heads, with head dimension 48. Reads and writes receive independent RMS normalization per head and rotary position encoding. Write/value heads are repeated across their four read-head groups, and causal attention is

$$\mathbf{y}_{t,h} = \sum_{j \leq t} \text{softmax}_j \left(\frac{\text{RoPE}(\mathbf{r}_{t,h})^\top \text{RoPE}(\mathbf{w}_{j,\kappa(h)})}{\sqrt{48}} \right) \mathbf{v}_{j,\kappa(h)}, \quad (2)$$

where $\kappa(h)$ maps each read head to one of the two shared write/value heads. PyTorch scaled dot-product attention is configured to permit FlashAttention dispatch on H200. W/R/V is therefore an attention mechanism: R and W are passed to the query and key boundary of SDPA, respectively, and V is the contextual value stream. The terminology describes data flow, not an attention-free operator. Incremental execution retains past W and V tensors and grows with prefix length. An exploratory lexical write residual exists in the implementation, but it is fixed to zero and skipped in every controlled run reported in Table 1.

2.2 Tied lexical residual object

The shared feed-forward transformation is a dense SwiGLU,

$$\mathbf{F}(\mathbf{h}) = \mathbf{W}_d [\text{SiLU}(\mathbf{W}_g \mathbf{h}) \odot \mathbf{W}_u \mathbf{h}]. \quad (3)$$

For token identity x_t , a tied table $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}| \times r}$ provides a low-rank modulation:

$$\mathbf{z}_t = \text{SiLU}(\mathbf{U}\mathbf{h}_t) \odot (\mathbf{1} + \mathbf{E}[x_t]), \quad (4)$$

$$\mathbf{O}(\mathbf{h}_t, x_t) = \mathbf{D}\mathbf{z}_t. \quad (5)$$

The same table $\mathbf{E}$ is shared across layers; projection matrices remain layer-specific. Setting $\mathbf{E}$ to zero makes the token-dependent modulation neutral initially, but the object branch itself remains active through a nonzero output gate α :

$$\mathbf{F}_{\text{object}}(\mathbf{h}_t, x_t) = \mathbf{F}(\mathbf{h}_t) + \alpha \mathbf{O}(\mathbf{h}_t, x_t). \quad (6)$$

2.3 Deterministic micro-expert residual

Each layer assigns token x_t to one of N micro-experts using

$$e_\ell(x_t) = (x_t + \ell) \bmod N. \quad (7)$$

All expert hidden activations are computed as one regular tensor operation and the selected output is gathered without sorting or scatter. With expert width w ,

$$\mathbf{R}_\ell(\mathbf{h}_t, x_t) = \beta \mathbf{W}_d^{(e_\ell(x_t))} \left[\text{SiLU}(\mathbf{W}_g^{(e_\ell(x_t))} \mathbf{h}_t) \odot \mathbf{W}_u^{(e_\ell(x_t))} \mathbf{h}_t \right]. \quad (8)$$

The complete feed-forward output is $\mathbf{F} + \alpha \mathbf{O} + \beta \mathbf{R}$. This is a lexical residual over a shared dense path, not a claim that routing replaces shared computation.

2.4 Fixed-state WVR associative ablation

The earlier fixed-state ablation uses one recurrent associative matrix rather than softmax attention. We denote that mechanism by *WVR*: write address, contextual value, and read address. For token x_t and hidden state $\mathbf{h}_t$, a compressed lexical forge reuses the tied lexical object table,

$$\mathbf{f}_t = \mathbf{W}_{f,2} \text{SiLU}(\mathbf{W}_{f,1} \mathbf{E}[x_t]), \quad (9)$$

where the implemented bottleneck is $16 \rightarrow 4 \rightarrow 128$. The common address combines this deterministic lexical code with a projected hidden-state signature,

$$\mathbf{c}_t = \text{norm}(\mathbf{W}_c \mathbf{h}_t), \quad (10)$$

$$\mathbf{a}_t = \text{norm}(\text{forge}(x_t) + \tanh(\eta) \mathbf{c}_t). \quad (11)$$

The lexical code is an address tied to token identity; it is not evidence of semantic representation. The contextual projection is likewise described operationally rather than semantically.

Writing uses the preceding address and the current contextual payload,

$$\mathbf{w}_t = \mathbf{a}_{t-1}, \quad \mathbf{v}_t = \mathbf{W}_v \mathbf{h}_t, \quad \mathbf{r}_t = \mathbf{a}_t, \quad (12)$$

$$\hat{\mathbf{v}}_t = \mathbf{w}_t^\top \mathbf{M}_{t-1}, \quad (13)$$

$$\mathbf{M}_t = \lambda \mathbf{M}_{t-1} + \rho \mathbf{w}_t (\mathbf{v}_t - \hat{\mathbf{v}}_t)^\top, \quad (14)$$

$$\mathbf{y}_t = \mathbf{r}_t^\top \mathbf{M}_{t-1}, \quad (15)$$

$$\mathbf{c}_t^{\text{WVR}} = \gamma \mathbf{W}_o \mathbf{y}_t. \quad (16)$$

The context residual $\mathbf{c}_t^{\text{WVR}}$ is added to the causal-convolution output. The implementation shifts the write stream causally: the current read uses $\mathbf{r}_t$, while the preceding write address is paired with the current payload. The chunkwise triangular solve is an exact parallelization of this recurrence. Thus WVR stores a contextual hidden value rather than a lexical code. The state is one rank-128 matrix plus fixed-size address and collision-load vectors, independent of prefix length.

Collision normalization maintains a decayed diagonal load estimate $\mathbf{d}_t$ and scales write and read coordinates by $(\mathbf{d}_t + \epsilon)^{-1/2}$ before normalization. It does not partition the rank-128 matrix into separate memories.

QKV attention compares a contextual query with every retained key and performs normalized competition over individual token occurrences. WVR instead reads the pre-update compressed recurrent association through $\mathbf{r}_t^\top \mathbf{M}_{t-1}$. Our training comparison concerns modeling quality and throughput without assigning credit for KV-cache size. The representational tradeoff is explicit: WVR has fixed state, but repeated or colliding occurrences may overwrite one another.

The experimental occurrence-address extension replaces the dense contextual projection by a compact $d \rightarrow 8 \rightarrow 128$ signature and adds a gated deterministic position signature,

$$\mathbf{a}_t = \text{norm}(\text{forge}(x_t) + \text{smallextx}(\mathbf{h}_t) + \tanh(\gamma) \text{pos}(t)), \quad (17)$$

with $\gamma = 0$ initially and one fixed-size position counter for incremental decoding. Its completed full-budget run is reported as a negative ablation.

2.5 Architectural path ablations

We retain three completed intermediate conditions: one W/R/V layer inserted into a collision-normalized WVR model; eight W/R/V layers with strongly lexical normalized addresses; and eight contextual W/R/V layers with two remaining WVR layers. These are path ablations, not isolated factorial controls. Their ordering motivates hypotheses about insufficient selective attention, overly dominant lexical addressing, and residual fixed-state bottlenecks, but does not identify those causes independently.

3 Experimental Design

3.1 Controlled comparison

The primary experiment compares dense GQA with contextual W/R/V attention. Both models use ten layers, hidden size 384, eight read/query heads, two shared write/value or key/value heads, and the same lexical-object micro-expert feed-forward path. GQA has 98,179,844 parameters and W/R/V has 98,195,204 (a 0.016% difference). The W/R/V lexical write residual is fixed to zero and skipped, while per-head RMS normalization of R and W remains active.

3.2 Data, tokenizer, and budget

All six controlled runs use the 200,019-entry `o200k_base` tokenizer and FineWeb-Edu `sample-10BT`, sequence length 2,048, BF16, batch size 16, and seeds 42–44. Each run performs 3,052 steps, or exactly 100,007,936 training tokens. A deterministic 5% document stream is reserved for evaluation and excluded from training. The final estimate averages 32 batches (1,048,576 tokens) and is taken at step 3,000. The archived dataset shard is pinned to revision 87f0914 with SHA-256 `b1ba7b2c...d78e871`.

Both conditions use AdamW, peak learning rate 4×10^{-4} , weight decay 0.1, identical warmup/decay, chunked exact-vocabulary loss, eager execution, and the same H200 software stack. Compilation is disabled for both because a tied lexical-table path produced non-finite gradients under the tested compiler stack; eager execution remained finite.

3.3 Metrics and throughput

We report held-out next-token cross-entropy and training throughput separately. Throughput for each run is the median of finite logged points from step 500 onward after excluding points below 80k tokens/s caused by evaluation, checkpoint, or network stalls. We report the mean of the three run medians. The paired loss difference is W/R/V minus GQA for the same seed. With $n = 3$, we provide the sample standard deviation and a two-sided 95% Student- t interval, but do not treat it as evidence of significance.

3.4 Archived evidence

Each controlled run has a realized JSON configuration, per-step CSV, final checkpoint, source commit, and SHA-256 manifest. The exact 2.15GB FineWeb-Edu Parquet shard is archived for offline reevaluation. Historical H100 WVR and W/R/V path runs use different protocols and appear only as descriptive development evidence; they are not pooled with H200 results.

3.5 Hardware and numerical checks

Controlled runs use one NVIDIA H200 with PyTorch SDPA permitting FlashAttention dispatch. FP32 unit tests cover W/R/V causality, padding masks, grouped head shapes, RoPE, cache behavior, fused R/W/V projection equivalence, lexical-off invariance, and full/incremental equivalence. The W/V cache grows with prefix length, like a grouped KV cache; W/R/V is not presented as fixed-state or attention-free.

4 Results

4.1 Controlled H200 comparison

W/R/V is lower in loss on all three seeds. The mean paired difference is -0.018604 , approximately 0.40% of the GQA mean loss. The two-sided 95% Student- t interval is $[-0.0541, 0.0169]$. Because it includes zero and contains only three pairs, we describe the ordering as consistent in this experiment rather than statistically established.

Table 1: Matched held-out loss by seed. Lower is better. Each value is the final step-3,000 evaluation over 1,048,576 held-out tokens.

Seed	GQA	W/R/V	W/R/V – GQA
42	4.683425	4.662531	−0.020894
43	4.717673	4.686053	−0.031620
44	4.708008	4.704711	−0.003297
Mean	4.703035	4.684432	−0.018604
Sample SD	0.017657	0.021137	0.014300

Table 2: Training throughput on one H200. Each seed contributes its post-warmup median after evaluation/checkpoint stalls are excluded; the table reports the mean and sample SD across seeds.

Mechanism	Mean train tok/s $\uparrow$	Sample SD
GQA	125,021	100
Contextual W/R/V	121,448	254

W/R/V is 3,573 tokens/s slower on average, a 2.86% throughput penalty. Quality and throughput therefore move in opposite directions: the controlled result is a modest quality improvement at modest training-speed cost, not a Pareto improvement.

4.2 Historical architecture path

Historical single-seed H100 runs motivated the controlled comparison but are not matched to the H200 protocol. Their key completed losses are 4.895427 for collision-normalized fixed-state WVR, 4.892026 after adding one W/R/V layer, 4.973798 for a strongly lexical eight-layer W/R/V condition, 4.852761 for an eight-layer contextual hybrid, and 4.706118 for the ten-layer contextual condition with RMS normalization. These values document the development path only. They are not combined with Table 1, and no causal credit is assigned from their joint changes.

Short proxies were useful failure screens but unreliable rankers: several favorable lexical-forge and multi-timescale proxy results did not improve at full budget. The final controlled model therefore removes the lexical write residual from the evaluated W/R/V attention path. Tied lexical objects and deterministic micro-experts remain in the shared feed-forward architecture of both GQA and W/R/V.

Two H200 seed-42 ablations are reported descriptively and are not pooled with the three-seed comparison. Enabling the lexical write residual gives 4.660939 loss versus 4.662531 with the residual disabled, a difference of −0.001592, while reducing median throughput from 121,479 to 119,121 tokens/s (−1.94%). Disabling per-head R/W RMS normalization gives 4.795811 loss, +0.133280 relative to the lexical-off normalized condition, at 123,257 tokens/s. The lexical-residual run uses an earlier source commit, and both comparisons have one seed; these observations motivate matched ablations but do not isolate causal effects.

4.3 Fixed-state negative controls

The historical fixed-state experiments remain informative negative controls. A separate 1B-token causal-convolution run cannot be ranked against the 100M-token conditions because its budget differs by a factor of ten. Archived associative-recall and induction probes report zero accuracy for evaluated attention-free checkpoints, but the available GQA probe does not document an identical protocol; no quantitative positive-control comparison is claimed.

FP32 unit tests reproduce full-sequence W/R/V outputs under token-by-token incremental execution at tolerance 10^{-5} . The retained W/V tensors grow with context, so these tests support causal implementation equivalence rather than a fixed-state serving claim.

5 Related Work

Transformers use contextual query–key competition and value aggregation (Vaswani et al., 2017). Grouped-query attention shares key/value heads across query groups to reduce memory and bandwidth (Ainslie et al., 2023). The central W/R/V operation is in this family: read heads play the query role, shared write heads play the key role, and values remain contextual. We do not claim that renaming streams creates a new attention operator. We instead evaluate the realized grouped R/W/V parameterization and normalization choices under a matched training protocol.

The final model also uses established components: rotary position encoding (Su et al., 2021), per-head read/write normalization closely related to query–key normalization (Henry et al., 2020), and exact IO-aware attention kernels (Dao et al., 2022). Our empirical question is whether this concrete combination changes the quality–throughput tradeoff under a small-model budget.

RWKV, RetNet, and Mamba develop recurrent or state-space alternatives with linear-time sequence processing (Peng et al., 2023; Sun et al., 2023; Gu & Dao, 2023). Our fixed-state WVR ablations are closer in motivation than the final W/R/V model, but they use a distinct single associative matrix and perform worse in the present controls. The final model abandons fixed-state decoding and retains grouped W/V caches; it should not be presented as a linear-state competitor to those methods.

6 Limitations and Broader Impact

The controlled evidence is limited to approximately 98.2M parameters, one tokenizer, one FineWeb-Edu shard, 100M training tokens, and three seeds. The paired ordering is consistent, but its 95% interval includes zero. It does not establish statistical significance, scaling behavior, downstream-task benefit, or universal superiority to GQA.

The comparison changes a bundle: the R/W/V parameterization and per-head R/W RMS normalization. The six-run evidence does not isolate normalization factorially. Exploratory single-seed lexical-write and no-normalization observations are excluded from the controlled table because they are not three-seed matched comparisons. Additional matched controls are required before assigning causal credit to one component.

Training throughput is implementation-, software-, and H200-specific. Eager execution was used because the tested compiler stack produced non-finite gradients in a shared lexical-table path. The reported 2.86% penalty is not a production inference estimate. W/R/V retains W/V tensors whose size grows with context, like a grouped KV cache. Long-context prefill, time to first token, serving latency distributions, memory bandwidth, and CUDA Graph replay remain outside the claims.

FineWeb-Edu inherits limitations of large web corpora, including uncertain provenance, representational imbalance, and potentially harmful content. The architecture does not remove these risks. Efficiency improvements can also lower the cost of deploying low-quality or harmful models. The archived dataset shard is included for exact reproducibility and should be handled under its source license and governance requirements.

Finally, the work is an architecture-path study rather than a claim of a novel softmax operator. R and W occupy the technical query/key boundary of SDPA. The value of the terminology is the explicit data-flow decomposition and executable comparison, not a claim that renaming Q/K/V alone is an algorithmic contribution.

7 Conclusion

We evaluated contextual W/R/V attention with eight read heads, two shared write/value heads, per-head R/W RMS normalization, RoPE, and causal SDPA. In six matched H200 runs, W/R/V achieved lower held-out loss than GQA on all three paired seeds: 4.684432 ± 0.021137 versus 4.703035 ± 0.017657 . Its mean training throughput was 121,448 versus 125,021 tokens/s, a 2.86% penalty.

The result is encouraging but deliberately narrow. Three pairs do not establish significance, the 95% interval includes zero, and the experiment does not isolate normalization from the broader parameterization. The contribution is a complete artifact-backed comparison—configs, CSVs, checkpoints, source commits, dataset shard, and checksums—that turns a historical architecture path into a reproducible hypothesis for larger-scale evaluation.

References

- Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, and Sumit Sanghai. Gqa: Training generalized multi-query transformer models from multi-head checkpoints. 2023.
- Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness. 2022.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. 2023.
- Alex Henry, Prudhvi Raj Dachapally, Shubham Pawar, and Yuxuan Chen. Query-key normalization for transformers. 2020.
- Bo Peng et al. Rwkv: Reinventing rnns for the transformer era. 2023.
- Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding. 2021.
- Yutao Sun et al. Retentive network: A successor to transformer for large language models. 2023.
- Ashish Vaswani et al. Attention is all you need. 2017.

A Reproducibility Checklist

The anonymous review package contains:

- a standalone mini-framework implementing contextual W/R/V, matched GQA, the shared lexical-object/micro-expert feed-forward path, training, evaluation, and targeted tests, with no dependency on the authors’ research framework;
- realized JSON configurations and per-step CSV metrics for GQA and W/R/V at seeds 42–44, plus the two exploratory seed-42 ablations;
- exact model parameter counts, token budgets, optimizer fields, numerical mode, backend information, and source commits;
- the pinned FineWeb-Edu revision, shard path, and SHA-256;
- a machine-readable summary from which Tables 1 and 2 are generated;
- unit tests for causality, grouped shapes, RoPE, lexical-off behavior, fused projection equivalence, and full/incremental cache behavior.

The archived shard is `sample/10BT/000_00000.parquet` from FineWeb-Edu revision `87f09149ef4734204d70ed1d046ddc9ca3f2b8f9`; its SHA-256 is `b1ba7b2ce4cb5ea6ef42dca40263eabb85f37700d01693a68e9b30a31d78e871`. Every run processes 32,768 tokens per step for 3,052 steps. Evaluation uses 32 deterministic held-out batches at step 3,000.

The submission ZIP excludes Git history, credentials, cloud addresses, hostnames, local user paths, transient logs, model checkpoints, and the 2.15GB dataset shard. Checkpoints are not required for the reported comparison; the pinned dataset identity permits exact retrieval. The final extracted ZIP and PDF metadata are scanned for identifying strings before submission.